

Mia (Zhijun) Qian

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EDUCATION

University of California, San Diego

La Jolla, CA

M.S. in Bioengineering (Computational Biology & Bioinformatics)

GPA: 3.9

Sep. 2025 - Expected June 2027

B.S. in Bioinformatics, Computer Science Minor

GPA: 3.6

Sep. 2021 - June 2025

RESEARCH EXPERIENCE

Nuno Bandeira Lab | UC San Diego Jacobs School of Engineering, Graduate Researcher

Sep. 2025 - Present

- Assessed UniMod-annotated modifications in large-scale mass spectrometry proteomics datasets by systematically validating peptide-spectrum matches based on charge state agreement, precursor mass accuracy, and b/y ion fragment support, and selected representative USIs as supporting evidence.
- Characterized observed mass offsets as candidate peptidoforms by comparing spectrum-derived delta masses with UniMod monoisotopic modification masses and resolving ambiguous site localization using spectrum-level constraints.
- Developed custom scripts to systematically enumerate combinations of known and candidate modifications within a defined mass tolerance, enabling the explanation of labile and previously unassigned mass shifts in mass spectrometry proteomics data.

Joseph Ecker Lab | Salk Institute for Biological Studies, Bioinformatics Programmer

Feb. 2025 – Dec. 2025

- Identified cluster-specific differentially methylated genes and chromatin-accessible regions from single-cell GpC methylation sequencing data of Arabidopsis using ALLCools in Python.
- Designed and automated scalable Bash pipelines to process large ALLC methylation datasets, call chromatin accessibility peaks with gNOMePeaks, and construct master peak sets across clusters using bedtools and UpSetR for cross-cluster comparison and visualization.

Neil Chi Lab | UC San Diego School of Medicine, Bioinformatics Programmer

June 2024 - June 2025

- Investigated differences in heart regeneration between terrestrial and aquatic axolotls under non-injury conditions by analyzing scRNA-seq heart datasets with R and Seurat. Conducted differential gene expression analysis with DESeq2.
- Performed bulk RNA-seq alignment and analysis across different cell lineages to compare the effectiveness of various differentiation protocols and identify uncovered gene program changes in MAB21L2 and RUNX1T1-KO cell lines.
- Constructed developmental trajectory for CVDC2-differentiated stem cells by analyzing human pluripotent stem cell scRNA-seq and snATAC-seq data with Seurat and Signac in R. Integrated data to reveal gene programs in cardiac development, cell differentiation, and epigenetic changes. Identified key cell-cell signaling pathways using CellChat.

Qingfei Jiang Lab | UC San Diego School of Medicine, Undergraduate Researcher

April 2023 - May 2024

- Performed RNA extraction, cDNA synthesis, and RT-qPCR on Jurkat, CUTTL1, and SUPT1 leukemia cell lines to compare ADAR1 isoform expression and examine their roles in cellular proliferation and migration.

PROJECTS

Proteomics-Based Prediction of Long-Term Weight Change

- Analyzed proteomic peptide-variant intensity and weight change data from a 52-week obesity study for use with supervised machine learning models to predict peptides associated with patients' weight gain.
- Preprocessed and Z-scored variant features, selected top predictors via ANOVA F-tests, trained Random Forest classifiers with stratified cross-validation, and visualized results using heatmaps and bipartite graphs.

Huffman Encoding & Decoding for Large Files

- Implemented a tree-based Huffman coding algorithm in C++ for compressing and decompressing large FASTA files, optimized file size while preserving data integrity for efficient storage and retrieval.

Differential Expression Analysis Tool

- Developed a Python and R-based tool for analyzing STAR and RSEM result files. Performed correlation checks and generated volcano plots to show differential expression from DESeq2.
- Enabled customizable data visualization, improving efficiency and reducing redundant computation.

SKILLS

Programming Languages: Python, C++, R, Java, Bash, MATLAB, HTML, CSS, JavaScript

Tools, Libraries & Technologies: Git, Linux, AWS, Conda, Jupyter Notebook, NumPy, Pandas, LaTeX, Figma

Bioinformatics: scRNA-seq, Bulk RNA-seq, snATAC-seq, SAMtools, STAR, Bowtie2, Peak Calling, Motif Finding, Multiple Sequence Alignment, Signac, Harmony, GO analysis, ggplot2, Seaborn, Matplotlib, CellChat, ALLCools, bedtools, UpSetR, gNOMePeaks, UniMod, UniProt